

A Nestin–Cyclin-Dependent Kinase 5–Dynamin-Related Protein 1 Axis Regulates Neural Stem/Progenitor Cell Stemness via a Metabolic Shift

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ABSTRACT

Neural stem/progenitor cells (NSPCs) transplantation provides an alternative approach for various central nervous system (CNS) diseases treatment, while the difficulties in NSPC acquisition and expansion limit their further application. Unveiling the mechanism of NSPC stemness regulation may contribute to its further application. Nestin, generally recognized as a marker of NSPCs, plays a crucial role in the CNS development and NSPC stemness maintenance. Here, we report that Nestin loss triggers mitochondrial network remodeling and enhances oxidative phosphorylation (OXPHOS) in NSPCs treated with Nestin RNA interference (RNAi). Mitochondrial morphology is dynamically controlled by the balance between fission and fusion mediators; one of these mediators, the pro-fission factor, dynamin-related protein 1 (Drp1), shows decreased activation in Nestin-knockdown cells. Upstream, Drp1 phosphorylation is under control of the cytosolic cyclin-dependent kinase 5 (Cdk5). Inhibition of Cdk5 using RNAi or a chemical inhibitor (roscovitine) induces mitochondrial elongation and promotes mitochondrial respiration, indicating that Cdk5-dependent Drp1 phosphorylation participates in mitochondrial metabolism and NSPC stemness regulation. Strikingly, Nestin knockdown results in Cdk5 redistribution, with less remaining in the cytosol, leading to mitochondrial remodeling. We identify Nestin1-640 sequesters Cdk5 in the cytosol and phosphorylates Drp1 subsequently. Together, our results show that a Nestin-Cdk5-Drp1 axis negatively regulates mitochondrial OXPHOS, which is indispensable for the maintenance of NSPC stemness. *STEM CELLS* 2018;36:589–601

SIGNIFICANCE STATEMENT

Nestin is considered as a common marker of multipotent stem cells, especially neural stem/progenitor cells (NSPCs). However, the mechanism through which Nestin contributes to NSPC maintenance is not fully understood. This study points out that Nestin–cyclin-dependent kinase 5–dynamin-related protein 1 axis promotes mitochondrial fission and glycolysis, which is indispensable for NSPC maintenance.

INTRODUCTION

Mammalian neural stem/progenitor cells (NSPCs) are undifferentiated precursors that retain the capacity to self-renew and generate various types of neurons and glia during central nervous system (CNS) development [1]. Under certain pathological conditions, the homing, recruitment, and transplantation of NSPCs could help the brain recover from various injuries, making NSPCs potential candidates for cell-based therapies [2, 3]. Although the functional properties of NSPCs have been extensively explored, the molecular mechanisms underlying their stemness maintenance remain incompletely understood.

Recent studies have demonstrated that metabolic plasticity plays an essential role in the pluripotency maintenance and lineage specification of pluripotent stem cells (PSCs) [4]. While PSCs largely rely on anaerobic glycolysis to sustain their pluripotent states, oxidative phosphorylation (OXPHOS) is the primary source of energy for cell differentiation [5]. For example, induced PSCs reprogramming by stemness-related transcription factors was shown to increase glycolytic enzyme expression and facilitate pluripotency-dependent glycolysis compared with that observed in differentiated cells [6]. Additionally, somatic stem cells such as hematopoietic stem cells (HSCs) have also been reported to

be dependent on their metabolic status regulation, as switching from glycolysis to OXPHOS impaired HSC function and promoted their differentiation potential [7]. Although metabolic plasticity has recently moved to the forefront of stem cell research [8–10], how metabolism participates in NSPC stemness maintenance and differentiation remains unclear.

Nestin is a class VI intermediate filament (IF) protein that was originally described in NSPCs of the CNS during embryonic development [11]. Considered as the marker of NSPCs, Nestin is downregulated upon the differentiation process [12]. Nestin is also expressed in other stem/progenitor cell populations, including mesenchymal stem cells and hair follicle stem cells [13, 14], suggesting that Nestin might be a common marker of multipotent stem cells [15]. Moreover, recent studies have shown the potential role of Nestin as more than a marker but also a dynamic signaling scaffold in cell fate determination [12]. Our previous research also demonstrated that Nestin could mediate the proliferation and invasion of gastrointestinal stromal tumors through its effect on ROS generation [16]. Here, we investigate whether Nestin expression in NSPCs could help maintain their stemness via its interaction with mitochondria and the underlying mechanisms through which it regulates mitochondrial metabolism.

MATERIALS AND METHODS

Isolation and Culture of NSPCs

Primary NSPCs were isolated from embryonic C57BL/6 mice cerebral cortices (E15.5d) [17]. C57BL/6 mice were purchased from the Animal Center of Sun Yat-sen University (Guangzhou, China). All animal studies were carried out in accordance with the guidelines of the Sun Yat-sen University Institutional Animal Care and Use Committee. Briefly, cortices were mechanically dissociated in ice-cold dissection buffer (Ca^{2+} - and Mg^{2+} -free Hank's balanced salt solution; Gibco, Waltham, MA, USA) and incubated in a buffer containing 0.025% trypsin (Gibco, Waltham, MA, USA) and 25 $\mu\text{g}/\text{ml}$ DNaseI (Sigma-Aldrich, St Louis, MO, USA) for 30 minutes at 37°C . The cells were washed and resuspended in growth medium comprising 1X N_2 supplement (Gibco, Waltham, MA, USA), 20 ng/ml epidermal growth factor (Gibco, Waltham, MA, USA), and 20 ng/ml basic fibroblast growth factor (bFGF; Gibco, Waltham, MA, USA) in Dulbecco's modified Eagle's medium (DMEM)/F12. The cells were cultured as floating neurospheres as described previously [18] or in multi-well chamber slides (Corning, Corning, NY, USA) coated with poly-D-lysine (PDL; 5 mg/ml, Sigma-Aldrich, St Louis, MO, USA) at 37°C in a humidified atmosphere containing 5% CO_2 . c17.2 mouse neural progenitor cells were purchased from European Collection of Cell Culture (ECACC) and cultured in DMEM (Hyclone, Logan, UT, USA) supplemented with 10% fetal bovine serum (FBS), 5% horse serum (Gibco, Waltham, MA, USA), 100 U/ml penicillin, and 100 $\mu\text{g}/\text{ml}$ streptomycin (Hyclone, Logan, UT, USA) at 37°C in a humidified atmosphere of 5% CO_2 .

Synchronization and Induction of Differentiation

For G_0 phase synchronization, c17.2 cells were maintained in medium supplemented with 0.1% FBS or 0.1% horse serum for 48 hours. Upon their transfer to medium containing 10% FBS, the cells re-entered the cell cycle.

Primary NSPC differentiation was induced by incubation in bFGF-free differentiation medium (DMEM/F12 containing 2% B27 supplement and 1% FBS) for 7 days followed after plating of single cell suspensions [19].

Vector Construction and Lentiviral Transduction

Gene expression was knocked down using synthetic shRNAs of the following sequences: *Nestin*: forward 5'CCGGGTGAGACTCTGGAATGCAATTCAAGAGATTGCATTCCAGAGTCTCACTTTTGG3' and reverse 5'AATTCAAAAAGTGAGACTCTGGAATGCAATTCAAGAGATTGCATTCCAGAGTCTCAC3'; and cyclin-dependent kinase 5 (Cdk5): forward 5'TGAAGTGTGTTCAAGGCTAAGTCTCTGTCATTAGCCTTGAACACAGTCTTTTTC3' and reverse 5'TCGAGAAAAAAGAACTGTGTTCAAGGCTAATGACAGGAAGTTAGCCTTGAACACAGTCA3'. The expressions of Nestin and Cdk5 in NSPCs were knocked down by transduction of lentiviral particles generated in 293FT cells. The knockdown efficiency was detected by immunoblotting and real-time polymerase chain reaction (RT-PCR) at 3 days post-transduction. dynamin-related protein 1 (Drp1)^{S592A} and Drp1^{S592E} mutant vectors were generated as shown in Supporting Information Figure S5, using the following primer sets: Drp1^{S592A}: forward 5'ATTCCAATTATGCCAGCCGCCACAAAAAGGTCATGCCGT3' and reverse 5'ACGGCATGACCTTTTGTGGGGCGGCTGGCATAATTGGAAT3'; and Drp1^{S592E}: forward 5'ATTCCAATTATGCCAGCCGAGCCACAAAAAGGTCATGCCGT3' and reverse 5'ACGGCATGACCTTTTGTGGCTCGGCTGGCATATTGGAAT3'. Expression plasmids for Drp1^{S592A} and Drp1^{S592E} were transfected using the X-treme GENE HP reagent (Roche Applied Science, Indianapolis, IN, USA) according to the manufacturer's instructions.

RNA Isolation and qRT-PCR

Gene expression was assessed by quantitative RT-PCR (qRT-PCR) as described previously [16]. Briefly, total RNA was extracted from cell lysates using the TRIzol reagent (Invitrogen, Carlsbad, CA, USA). First-strand cDNA was synthesized with a QuantiTect Reverse Transcription kit (Qiagen, Germantown, MD, USA) according to the manufacturer's instructions, and qRT-PCR was performed with the SYBR Green qPCR SuperMix (Roche Applied Science, Indianapolis, IN, USA) and a Light Cycler 480 Detection System (Roche Applied Science, Indianapolis, IN, USA). The level of each target mRNA was normalized with respect to that of β -actin. The sequences of the primers used for qRT-PCR are listed in Supporting Information Table S1.

Western Blotting

For Western blotting, NSPCs were collected, washed twice with cold phosphate-buffered saline (PBS), directly lysed in 1X RIPA buffer, and then centrifuged at 15,000g for 5 minutes at 4°C . Each supernatant was recovered as a total cell lysate. Equal amounts of protein were resolved by SDS-polyacrylamide gel electrophoresis and then electrotransferred to a 0.45- μm pore-sized polyvinylidene difluoride membrane (Millipore, Darmstadt, Germany). Specifically bound primary antibodies were detected using horseradish peroxidase-coupled secondary antibodies and enhanced chemiluminescence (Millipore, Darmstadt, Germany). The primary and secondary antibodies used are listed in Supporting Information Table S2.

Immunofluorescence

For immunofluorescence, cells were attached to PDL-coated glass slides, fixed for 20 minutes with 4% paraformaldehyde, permeabilized in 0.1% Triton X-100 for 10 minutes, sequentially incubated with primary and secondary antibodies (listed in Supporting Information Table 2), and subjected to nuclear staining with 4',6-diamidino-2-phenylindole. Images were acquired at room temperature using an LSM780 confocal microscope (Zeiss, Pleasanton, CA, USA).

Mitochondrial Morphology Analysis

Mitochondria were defined as follows: tubular, interconnected and branched mitochondrial networks with length > 5 μm ; intermediate, length between 2.5 and 5 μm ; and fragmented, length of < 2.5 μm .

Clonal Neurosphere-Formation Assay

Neurospheres were dissociated into a single-cell suspension and reseeded into ultralow-binding six-well plates at 100 cells per square centimeter. Clonally derived neurospheres were counted and their diameters were measured using Image J software.

Isolation of Mitochondrial Fractions

Mitochondrial fractionation was performed using a Cell Mitochondria Isolation kit (Beyotime Biotechnology, Shanghai, China) according to the manufacturer's instructions. Briefly, cells were harvested and lysed in a hypotonic solution for 15 minutes at 4°C. The nuclei were collected, the supernatant was centrifuged at 1,300g for 10 minutes at 4°C, the supernatant was centrifuged at 15,000g for 20 minutes, and soluble cytosolic and membrane fractions were collected.

Measurement of Mitochondrial Oxygen Consumption Rate and Extracellular Acidification Rate

The oxygen consumption rate (OCR) and extracellular acidification rate (ECAR) levels were both determined using a Seahorse Bioscience Extracellular Flux Analyzer (XF24; Seahorse Bioscience, Billerica, MA, USA) as described previously [20]. Briefly, cells were seeded to the XF24 cell culture plate and allowed to attach for 24 hours. The cell culture medium was replaced with XF medium, and the cells were incubated in a CO₂-free atmosphere for 1 hour. The basal OCR was measured in untreated control cells, and the maximal OCR was measured in cells treated with oligomycin (ATP synthase inhibitor; 1 μM) and carbonyl cyanide-p-trifluoromethoxyphenylhydrazone (FCCP, electron transport chain accelerator; 0.5 μM). Finally, 1 μM Antimycin A was used to shut down all mitochondrial respiration and the OCR was tested. Each measurement of OCR or ECAR was repeated in triplicate.

Mitochondrial Membrane Potential Assessment

The mitochondrial membrane potential (MMP) levels were assessed by incubating cells for 20 minutes with Mitotracker Reagent (Life Technologies, Carlsbad, CA, USA) at 37°C. Stained cells were washed with PBS, and Mitotracker fluorescence was analyzed by flow cytometry.

Metabolism Assay

Intracellular ATP production and NAD⁺/NADH levels were measured using a CellTiter-Glo Luminescent Cell Viability Assay kit

(Promega, Madison, WI, USA) and a NAD⁺/NADH Assay Kit (BioVision, Milpitas, CA, USA), respectively, per the manufacturers' instructions. Extracellular lactate production and intracellular glucose uptake were measured using lactate and glucose assay kits (BioVision, Milpitas, CA, USA), respectively.

Kinase Assay

For kinase assays of whole cell and cytosolic fraction, cells (2×10^7 per sample) were lysed with 600 μl hypotonic buffer for 30 minutes on ice and then centrifuged at 3000 rpm for 10 minutes at 4°C. The supernatant was collected as the cytosolic fraction and the remaining portion was obtained as the nuclear fraction. The nuclear fraction was further lysed with 100 μl immunoprecipitation (IP) buffer (25 mM HEPES, pH 7.4, 150 mM NaCl, 5 mM EDTA, 0.5% Triton X-100, 1 mM phenylmethylsulfonyl fluoride, 1 mM dithiothreitol) for 30 minutes. The lysates were purified by centrifugation at 13,000 rpm, and the supernatant was obtained. The cytosolic fraction and nuclear lysates were prepared using protein G Sepharose beads (Amersham Biosciences, Piscataway, NJ, USA), followed by Cdk5 IP with anti-Cdk5 antibody, respectively. The immunocomplexes were harvested by centrifugation and washed three times with kinase buffer (Cell Signaling Technology, Danvers, MA, USA). The protein G Sepharose beads were resuspended with 20 μl of kinase buffer containing 125 μM cold ATP and H1 (100 $\mu\text{g}/\text{ml}$) to initiate the reaction. The reaction was allowed to proceed for 10 minutes, and then stopped by boiling for 5 minutes. Finally, the samples were analyzed by Western blotting performed as described above.

Statistical Analysis

All data are presented as the mean $\pm$ SD from at least three independent experiments. *p* values were analyzed using the Student's *t* test. The statistical package R (version 3.1.1) was used for all analyses. A two-sided *p* < .05 was considered significant.

RESULTS

Nestin Knockdown Induces Mitochondrial Elongation and Attenuates Self-Renewal in NSPCs

To identify the potential role of Nestin in mitochondrial dynamics and function regulation, we constructed lentiviral shNestin vectors, shNestin-1 and shNestin-2, and transfected them into primary NSPCs isolated from embryonic day 15.5 (E15.5) mouse cortices. We observed that the interference efficiency of shNestin-2 was relatively lower than that of shNestin-1 (Supporting Information Fig. S1A, S1B), so we used the latter sequence in our subsequent experiments.

We first examined the self-renewal and differentiation characteristics of NSPCs after transfection with shNestin-1. Primary NSPC proliferation was substantially reduced after Nestin knockdown by Ki67 staining (Fig. 1A). Since the number and diameter of neurospheres provide reliable estimates of the self-renewal and proliferation capacities of NSPCs [21], we measured the neurosphere formation capacity of Nestin-knockdown and non-targeting control (NTC) cells. The results showed that forced downregulation of Nestin dramatically decreased the number and diameter of neurospheres in NSPC cultures (Supporting Information Fig. S1C, S1D). Additionally, stemness markers including Sox2, Pax6, Hes1, and Hes5 were found

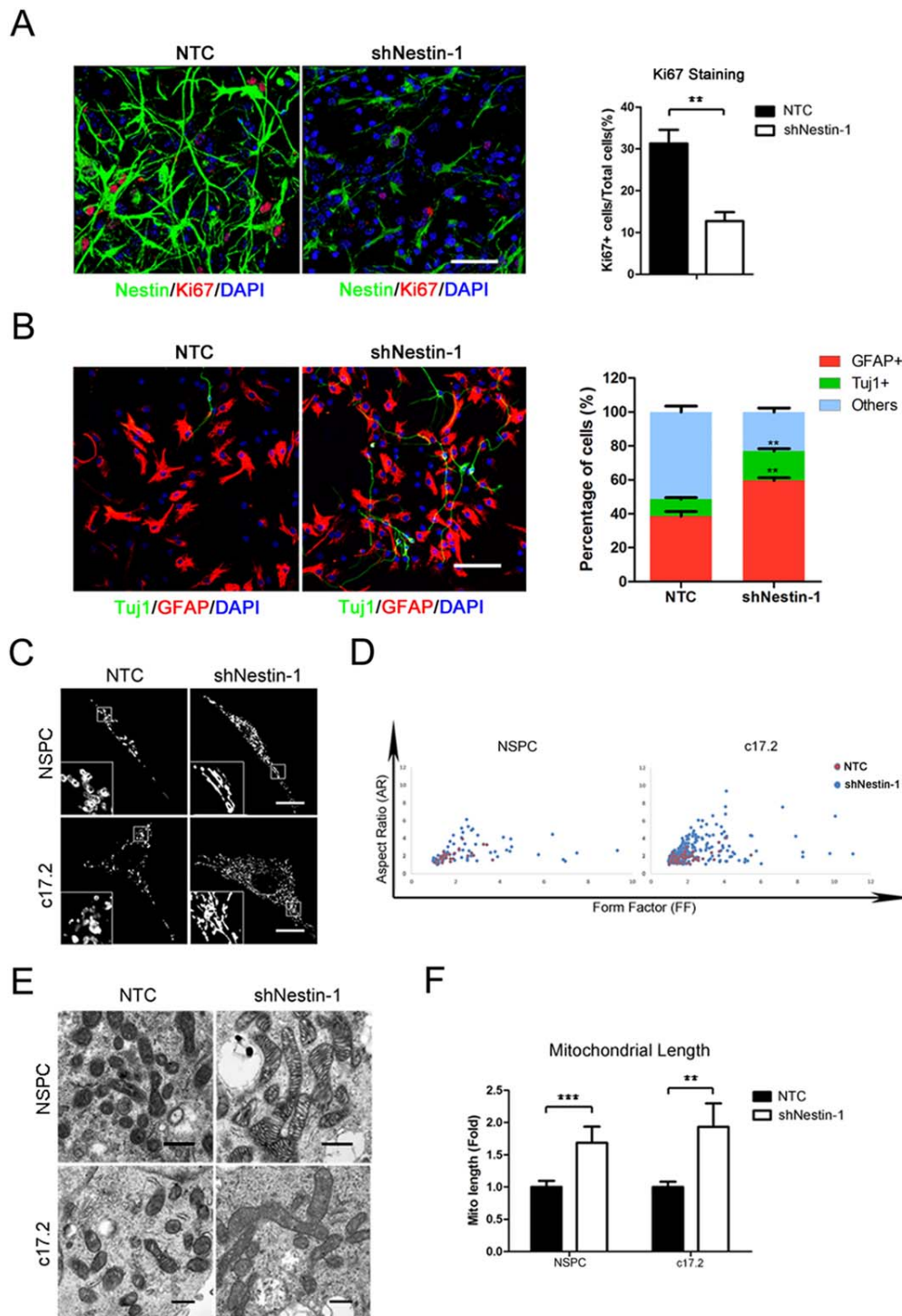


Figure 1. Nestin knockdown attenuates self-renewal and induces mitochondrial elongation in NSPCs. **(A):** Primary NSPCs were transfected with a shNestin-expressing vector and co-stained with anti-Nestin (green) and anti-Ki67 (red) antibodies. Representative confocal images are shown on the left. Scale bar = 100 μ m. Ki67+ cells were counted from five ($n = 5$) independent low-magnification microscopic fields (right). **(B):** Number of glial fibrillary acidic protein positive astrocytes (red) and Tuj1+ neurons (green) generated by adherent NSPCs cultured in differentiation medium for 5 days ($n = 4$). Scale bar = 100 μ m. **(C):** Immunofluorescent staining of mitochondria in NTC and Nestin-knockdown NSPCs was performed using an anti-COX IV antibody. Scale bar = 10 μ m. **(D):** Mitochondrial morphology ($n = 10$) was compared between NTC cells and Nestin-knockdown cells by calculating the form factor (FF) and aspect ratio (AR) values of representative confocal images. **(E):** Transmission electron microscopic images showed that Nestin-knockdown cells ($n = 96$ mitochondria) contained a higher proportion of elongated mitochondria than NTC cells ($n = 94$ mitochondria). Scale bars = 500 nm. **(F):** The mitochondrial length was quantified using the Image J software. All data are expressed as means $\pm$ SD; **, $p < .01$; ***, $p < .001$ by t test. Abbreviations: DAPI, 4',6-diamidino-2-phenylindole; GFAP, glial fibrillary acidic protein; NSPC, neural stem/progenitor cell; NTC, non-targeting control.

downregulated, while neuronal genes (Tuj1 and NeuN) and glial gene (glial fibrillary acidic protein [GFAP], S100B) were upregulated after Nestin knockdown (Supporting Information Fig. S1E). Furthermore, to test the impact of Nestin knockdown on NSPC differentiation, NSPCs were subjected to *in vitro* differentiation for 5 days. We found that there was no significant difference in survival rates of the differentiated cells between NTC and Nestin-knockdown groups, but the percentages of GFAP-positive astrocytes and Tuj1-positive neurons were higher in Nestin-knockdown cells than in NTC cells (Supporting Information Fig. S2A; Fig. 1B). Interestingly, the differentiation rates into glia and neurons were similar at 14 days of late stage (Supporting Information Fig. S2B). Similar trends of GFAP and Tuj1 mRNA expression were also observed at early and late phases of differentiation induction (Supporting Information Fig. S2C). Together, these data show that Nestin is crucial for NSPC stemness maintenance and that Nestin depletion accelerates the neuronal and glial differentiation of NSPCs. We then studied the mitochondrial morphology alterations after Nestin knockdown. The mitochondrial networks were elongated in Nestin-knockdown cells when compared with those in NTC cells by immunofluorescence staining (Fig. 1C, 1D; Supporting Information Fig. S3A). In addition, the electron microscopy results demonstrated that the majority of mitochondria in NSPCs were swollen, circular, and disorganized with sparse and irregular cristae, whereas those of Nestin-knockdown cells were elongated and linear with well-stacked cristae (Fig. 1E, 1F). Recent studies have demonstrated that cells undergo mitochondrial morphological changes during the cell cycle [22, 23]. To distinguish the effect of Nestin on mitochondria during the cell cycle, we synchronized the NTC and Nestin-knockdown c17.2 cells using serum starvation and then detected their mitochondrial morphologies at different time points. NTC cells underwent the expected mitochondrial morphological changes during the cell cycle. The mitochondrial length was generally much longer in Nestin-knockdown cells than in NTC cells, regardless of the cell cycle phase (Supporting Information Fig. S3B, S3C). Moreover, the expression levels of both inner (TIM23) and outer (TOM40) mitochondrial membrane translocators, usually reflect the mitochondrial mass [24], were expressed at similar levels after Nestin knockdown, suggesting that the effect of Nestin-knockdown is primarily on mitochondrial remodeling but not the total amounts of mitochondria (Supporting Information Fig. S4). In sum, these morphological assessments show that the mitochondrial length increased in Nestin-knockdown cells relative to NTC cells.

Nestin Knockdown Enhances Mitochondrial Function and Alters the Metabolic State

Mitochondria undergo fusion and fission dynamic states, which contribute to the metabolic and other functional changes in mitochondria [25]. Therefore, we evaluated the impact of Nestin knockdown on the mitochondrial respiratory function of NSPCs and measured the OCR using an extracellular flux analyzer. Nestin-knockdown cells displayed an increase in basal and maximal oxygen consumption compared with that in NTC cells (Fig. 2A), indicating an increase in mitochondrial respiration. Repressed glycolysis in Nestin-knockdown cells was reflected by a marked reduction in ECAR levels (Fig. 2B), whereas there was no significant difference in glucose uptake between Nestin-knockdown and NTC cells (Fig. 2C). Mitochondria use oxygen to

create ATP with by-products including ROS and nicotinamide adenine dinucleotide (NADH) via the OXPHOS machinery [7, 26, 27]. Thus, we next determined the mitochondrial NADH and ATP production in Nestin-knockdown and NTC cells and found that NADH and cellular ATP were increased, while the NAD⁺/NADH ratio was decreased in Nestin-knockdown cells relative to that in NTC cells (Fig. 2D, 2E). Furthermore, Nestin knockdown also increased the MMP in NSPCs (Fig. 2F), indicating that the mitochondrial function was enhanced in Nestin-knockdown cells. Together, these data suggest that the mitochondria exist in a hyperenergetic state upon Nestin downregulation, while those in NSPCs prefer glycolysis.

Drp1 Serine 592 Phosphorylation Governs Mitochondrial Morphology and Stemness in NSPCs

The morphology of the mitochondrial network reflects a competitive balance between the machineries responsible for mitochondrial fission and fusion [28]. Drp1 is the central mediator of mitochondrial fission, whereas mitofusin 1 (MFN1), MFN2, and optic atrophy 1 (OPA1) are required for mitochondrial fusion [29]. We failed to detect any change in the expression levels of these central mediators in Nestin-knockdown cells relative to that in NTC cells using qRT-PCR (Fig. 3A). Besides that, these mediators also showed no significant alterations in protein expression levels, except for the slight downregulation of MFN2 (Fig. 3B). Since MFN2 is widely accepted responsible for mitochondrial fusion, its downregulation is supposed to contribute to mitochondrial fragmentation instead of the fusion phenotype in Nestin-knockdown cells. These results indicated other possibilities for the mitochondrial fusion after Nestin knockdown. A recent study demonstrated that the phosphorylation of Drp1 at Ser592 could activate Drp1 and promote mitochondrial division [30]. We tested the phosphorylation levels of Drp1 at Ser592 in the NTC and Nestin-knockdown cells. As shown in Figure 3C, the activation of Drp1 at serine 592 was markedly decreased in Nestin-knockdown cells. To further determine the role of phosphorylated Drp1 in the dynamic alterations of mitochondria in Nestin-knockdown cells, we constructed a gain-of-function Drp1^{S592E} vector (phospho-mimetic) mutation and a loss-of-function Drp1^{S592A} vector (phospho-inhibitory) mutation (Supporting Information Fig. S5A, S5B) and examined the effects of exogenous Drp1^{S592E} and Drp1^{S592A} in the NTC and Nestin-knockdown cells. In NTC cells, Drp1^{S592E} expression was associated with a higher degree of mitochondrial fragmentation, whereas Drp1^{S592A} contributed to mitochondrial elongation (Fig. 3D, 3E). In Nestin-knockdown cells, Drp1^{S592E} expression was sufficient to reverse the changes in mitochondrial function (reflected by MMP) and metabolism (NADH generation, ATP production, and OCR) induced by Nestin depletion (Fig. 3F, 3G; Supporting Information Fig. S6), indicating that Drp1 phosphorylation dominates the mitochondrial regulation of NSPCs. Moreover, the clonal-formation defects of Nestin-knockdown cells were partially rescued by exogenous expression of Drp1^{S592E} (Fig. 3H, 3I). Taken together, these results suggest that hyperactivated Drp1 regulates mitochondrial fission, reduces mitochondrial respiration, which may facilitate the NSPC maintenance.

Cdk5 Activates pDrp1^{S592} in NSPCs

Previous studies have shown that Cdk family kinases, especially Cdk1 and Cdk5, could phosphorylate Drp1 and enhance

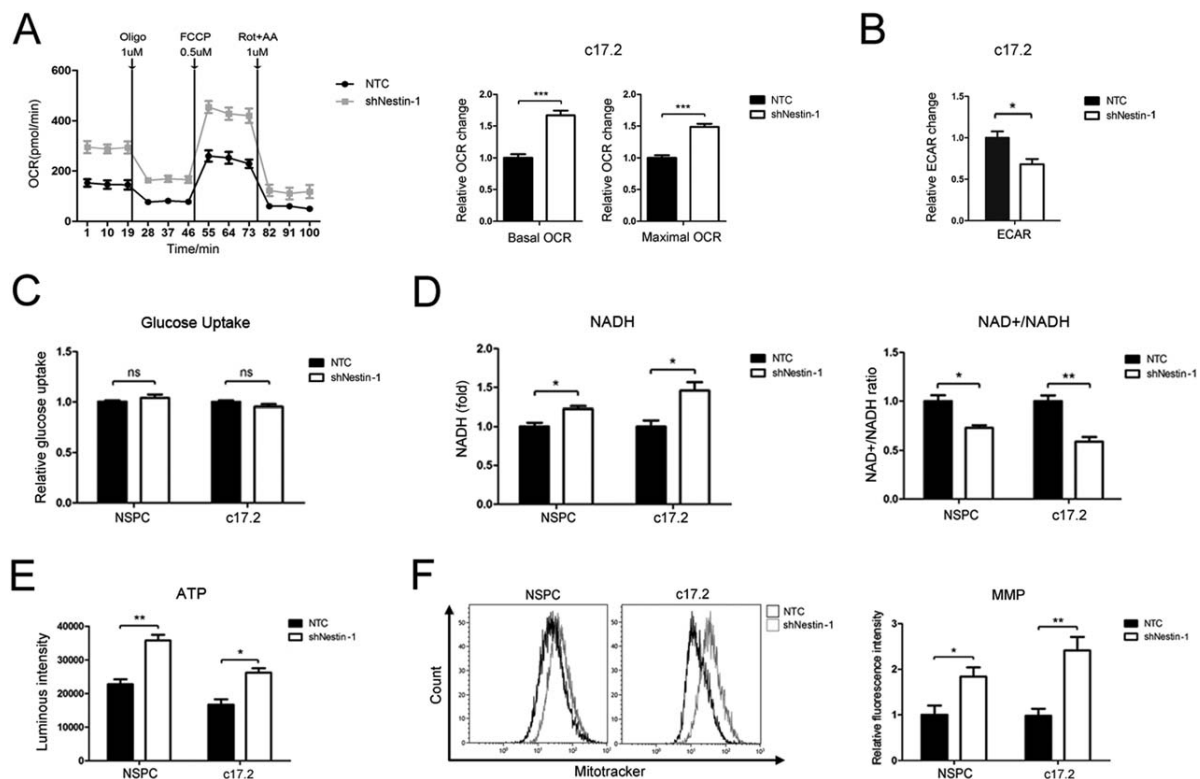


Figure 2. Nestin knockdown enhances mitochondrial function and alters the mitochondrial metabolic phenotype. **(A):** Nestin knockdown increases mitochondrial respiration, as measured using a Seahorse XF24 analyzer (left). The basal and maximal oxygen consumption rate ($n = 3$) are presented to the right. **(B, C):** Extracellular acidification rate (B) and glucose uptake levels (C) were determined after Nestin knockdown ($n = 3$). **(D, E):** In lentivirus (shNestin)-transduced cells, the NAD⁺/NADH ratio was decreased (D), while the ATP production was enhanced (E). **(F):** Density plot analysis ($n = 3$) of mitochondrial membrane potential levels in non-targeting control and shNestin cells, as determined by flow cytometry. All data are expressed as means $\pm$ SD; *, $p < .05$; **, $p < .01$; ***, $p < .001$, as assessed by t test. Abbreviations: ECAR, extracellular acidification rate; FCCP, carbonyl cyanide-*p*-trifluoromethoxyphenylhydrazone; MMP, mitochondrial membrane potential; ns, nonsignificant; NSPC, neural stem/progenitor cell; NTC, non-targeting control; OCR, oxygen consumption rate.

its activity to induce mitochondrial fission [31–33]. To investigate the molecular mechanism responsible for activating Drp1 in NSPCs, we assessed Drp1 phosphorylation and mitochondrial morphology in cells treated with roscovitine (a pan-Cdk1/2/5 inhibitor) or BMS-265246 (a Cdk inhibitor specific for Cdk1/2). We found that roscovitine treatment clearly reduced Drp1^{S592} phosphorylation, whereas BMS-265246 did not alter this phosphorylation status (Fig. 4A). Based on this finding, we examined the functional role of Cdk5 in mitochondrial regulation and found that roscovitine-treated cells exhibited more elongated mitochondria (Fig. 4B, 4C), repressed lactate generation, and enhanced mitochondrial OXPHOS and MMP (Supporting Information Fig. S7). Treatment with BMS-265246 did not significantly impact any mitochondrial dynamics or function (Fig. 4A–4C; Supporting Information Fig. S7). These results show that Cdk5 plays a dominant role in the ability of Drp1^{S592} to mediate phosphorylation and regulate mitochondrial dynamics in NSPCs.

Cdk5-knockdown and -overexpression vectors were constructed to further confirm this hypothesis. RNA interference (RNAi)-mediated Cdk5 knockdown specifically diminished Drp1^{S592} phosphorylation (Fig. 4D) and reduced the proportion of fragmented mitochondria by blocking Drp1 activity, whereas Cdk5 overexpression promoted Drp1 activation and increased the proportion of mitochondria fragmentation

(Supporting Information Fig. S8A, S8B). In terms of mitochondrial functions, Cdk5 knockdown inhibited lactate production and increased the production of ATP and NADH (Fig. 4E–4G), whereas Cdk5 overexpression slightly increased lactate production and suppressed mitochondrial respiration (Supporting Information Fig. S8C–S8E). For stemness maintenance, Cdk5 knockdown was also found repressed NSPCs self-renewal (Fig. 4H, 4I). These results strongly suggest that Cdk5 may act as an upstream driver of mitochondrial fission and metabolic control in NSPCs by phosphorylating Drp1 at residue Ser592.

Nestin Interacts with Cdk5 and Inhibits Its Translocation into the Nucleus

Our above results demonstrated that the elongated mitochondria in Nestin-knockdown cells resulted from the loss of activated Drp1, which was under Cdk5 regulation. As emerging evidence has indicated that IF networks serve as scaffolds for the organization of intracellular kinases [34], we used Nestin RNAi to investigate whether there was an underlying link between Nestin and Cdk5 in NSPCs. Nestin knockdown did not affect Cdk5 expression in whole-cell lysates (Fig. 5A). However, immunofluorescence confocal microscopy (Fig. 5B) and Western blotting analysis of specific cell fractions (Fig. 5C) detected more Cdk5 in the nuclei of Nestin-knockdown cells than NTC cells. These findings show that the total levels

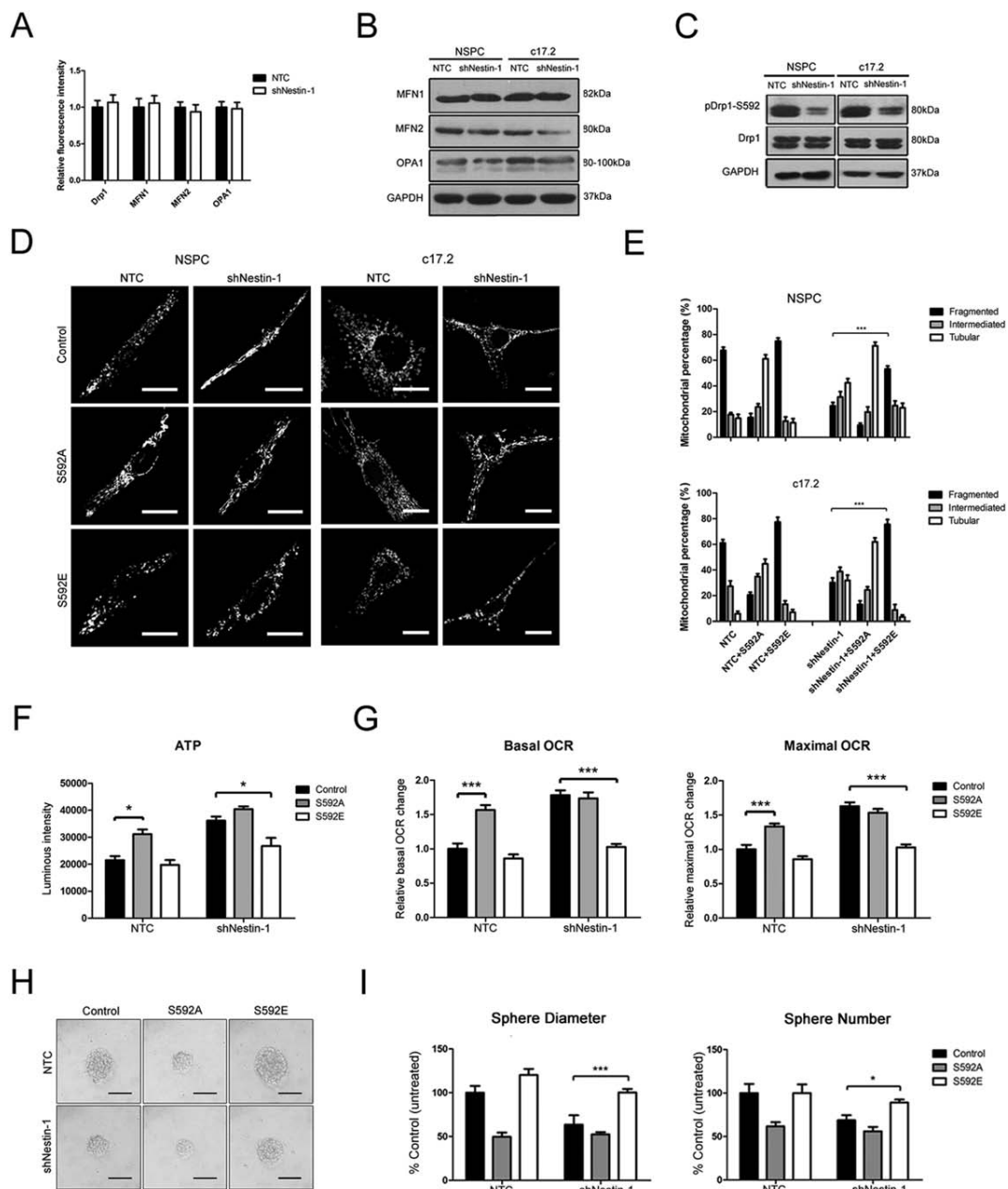


Figure 3. Drp1 Serine 592 phosphorylation governs mitochondrial morphology and stemness in NSPCs. **(A, B):** Quantitative real-time polymerase chain reaction (A) and immunoblotting analysis (B) of the mitochondrial fission/fusion mediators. **(C):** Western blotting analysis of total Drp1 and phosphorylated (activated) Drp1 (pDrp1^{S592}) in NTC and Nestin-knockdown cells. **(D):** NTC and Nestin-knockdown cells were subjected to exogenous transduction of pDrp1^{S592E} or pDrp1^{S592A}, and mitochondrial morphology was detected using an anti-COX IV antibody. Scale bar = 10 μ m. **(E):** Quantification analysis of mitochondrial morphology. Ten images ($n = 10$) and more than 100 cells were used for each group, respectively. **(F):** Alterations in ATP content ($n = 3$) paralleled the changes in the oxygen consumption rate (OCR) among cells expressing pDrp1^{S592E} and pDrp1^{S592A}. **(G):** Transfection of vectors encoding pDrp1^{S592E} diminished the OCR in both NTC and Nestin-knockdown cells, whereas expression of pDrp1^{S592A} increased the OCR in NTC cells, but not Nestin-knockdown cells. **(H):** Bright-field images ($n = 3$) of neurospheres formed from NSPCs under low-density culture. Scale bar = 100 μ m. **(I):** The average neurosphere diameters and numbers are presented as means $\pm$ SD; *, $p < .05$; ***, $p < .001$, as assessed by t test. Abbreviations: Drp1, dynamin-related protein 1; MFN, mitofusin; NSPC, neural stem/progenitor cell; NTC, non-targeting control; OCR, oxygen consumption rate.

of Cdk5 do not change dramatically after Nestin knockdown but the nuclear/cytoplasmic ratio increases, with less Cdk5 remaining in the cytosol. The comparison of cytosolic kinase activity versus total kinase activity revealed that cytosolic Cdk5 activity was markedly decreased in Nestin-knockdown cells (Fig. 5D), suggesting that the change in the subcellular

distribution of Cdk5 was paralleled by a change in its activity. Furthermore, the Cdk5 expression in Nestin-positive NSPCs of the embryonic brain (E15.5) was almost exclusively distributed in the cytosol, whereas the Cdk5 staining of Nestin-negative cells in the cortex area was generally accumulated in the nucleus (Fig. 5E; Supporting Information Fig. S9). These results

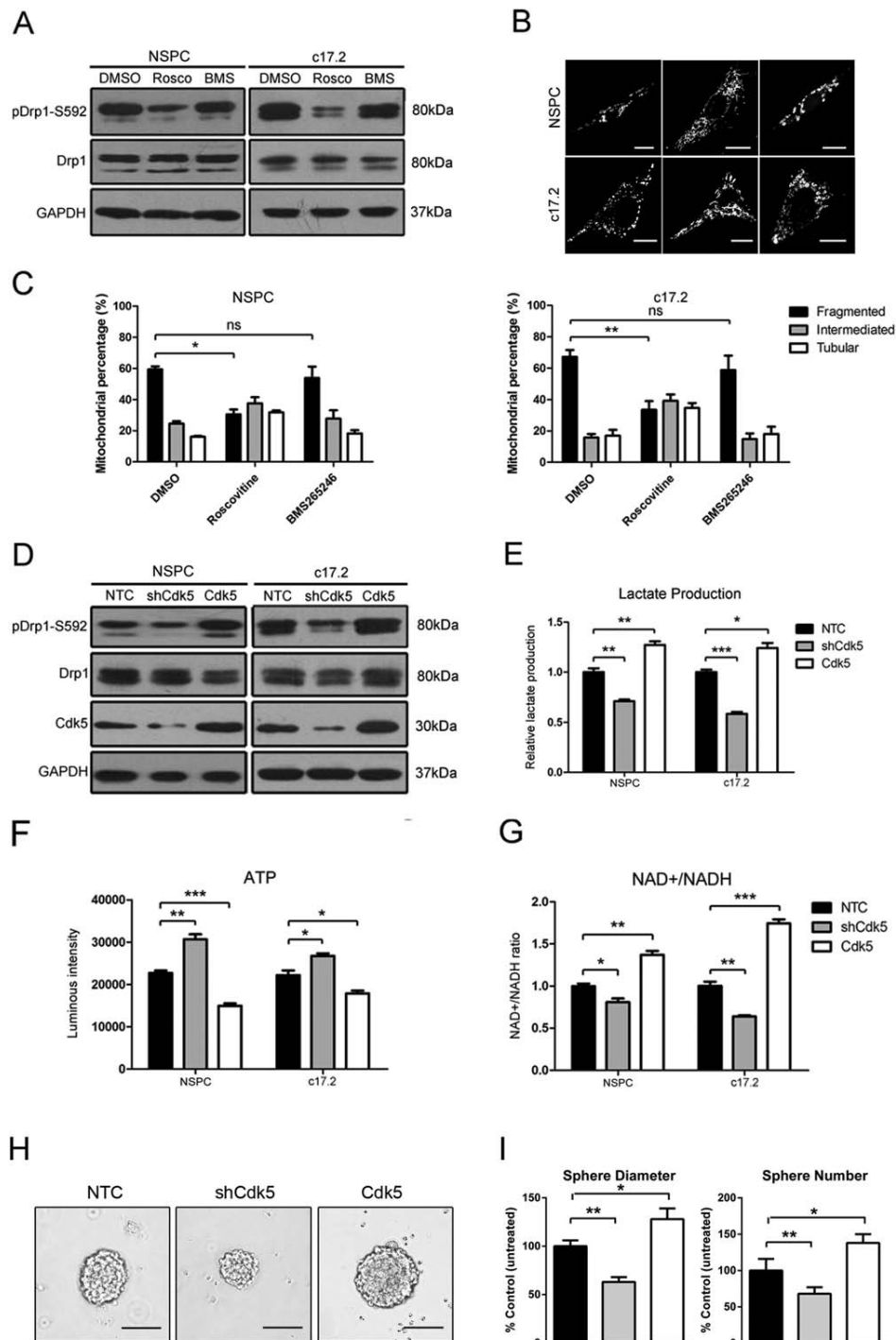


Figure 4. Cdk5 activates phosphorylated (activated) pDrp1^{S592} in NSPCs. **(A):** Lysates of non-targeting control and Nestin-knockdown cells were treated with roscovitine (10 μ M), BMS-265246 (1 μ M), or dimethyl sulfoxide, and then subjected to immunoblot analysis with the indicated antibodies against Drp1 and pDrp1^{S592}. **(B):** The treatment-induced changes in mitochondrial morphology were determined by immunofluorescent staining of the mitochondrial marker, COX IV. Scale bar = 10 μ m. Ten images ($n = 10$) and more than 100 cells were used for each group, respectively. **(C):** Roscovitine treatment promoted mitochondrial elongation in both primary NSPCs and c17.2 cells. **(D):** Drp1 phosphorylation was diminished in NSPCs and c17.2 cells transfected with shCdk5. **(E):** The lactate content was decreased in Cdk5-knockdown cells ($n = 3$). **(F, G):** Cdk5 overexpression impaired the productions of ATP (F) and NADH (G). **(H):** The impact of Cdk5 expression on NSPCs stemness was determined using neurosphere forming assay. **(I):** Neurosphere diameter and number were significantly decreased upon Cdk5 knockdown. All data are expressed as means $\pm$ SD; *, $p < .05$; **, $p < .01$; ***, $p < .001$, as assessed by *t* test. Abbreviations: Cdk5, cyclin-dependent kinase 5; DMSO, dimethyl sulfoxide; Drp1, dynamin-related protein 1; ns, non-significant; NSPC, neural stem/progenitor cell; NTC, non-targeting control; (pDrp1)S592, phosphorylated (activated) dynamin-related protein 1.

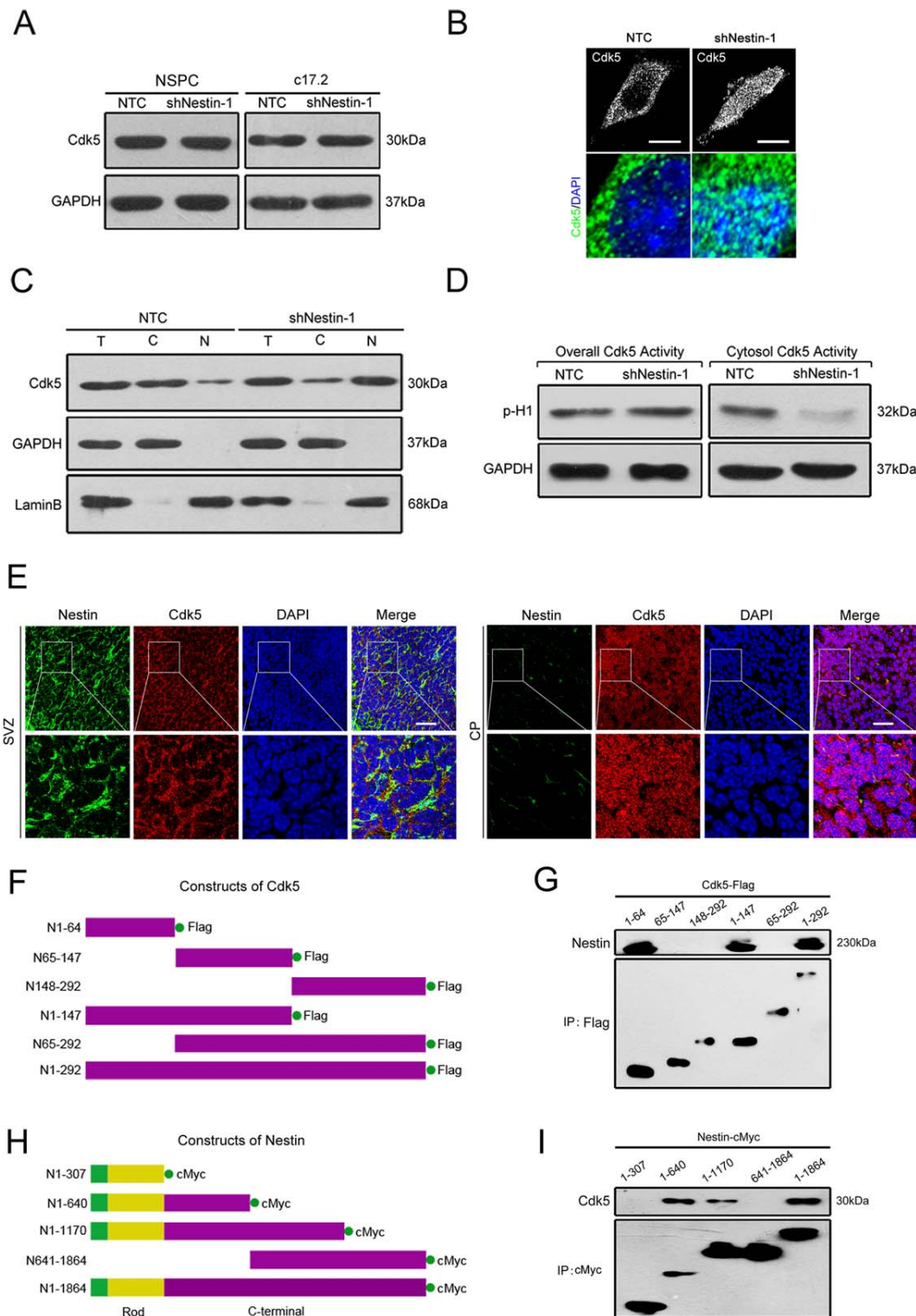


Figure 5. Nestin interacts with Cdk5 and inhibits its translocation into the nucleus. **(A):** Immunoblotting was used to measure Cdk5 protein levels in NTC and Nestin-knockdown cells. **(B):** Immunofluorescent staining showed that Cdk5 was redistributed in Nestin-depleted cells. Scale bar = 10 μ m. **(C):** Western blotting was used to detect the levels of Cdk5 in cytoplasmic (C) and nuclear (N) fractions of NSPCs. GAPDH and lamin B served as cytoplasmic and nuclear loading controls, respectively. **(D):** Kinase assays were used to assess total and cytosolic Cdk5 activity in NTC and Nestin-knockdown cells. **(E):** Representative immunofluorescent staining of Nestin and Cdk5 in the primary embryonic brain tissue (E15.5d). Scale bar = 50 μ m. **(F):** Schematic diagram of Cdk5 constructs. Flag-tagged Cdk5 plasmids were denoted by the corresponding amino acid sequences. All Cdk5 fragments express as a Flag fusion protein. **(G):** Nestin was immunoprecipitated from c17.2 cells expressing Flag-tagged Cdk5 constructs. Only Cdk5N1-64, Cdk5N1-147, and Cdk5N1-292 were found coimmunoprecipitated with Flag, indicating the Nestin binding area as Cdk5N1-64. **(H):** The schematic diagram of Nestin constructs, among which NestinN1-307 represented the rod domain in Nestin. The C-terminal was also separated by similar length. All Nestin fragments express as a c-Myc fusion protein. **(I):** Cdk5 was immunoprecipitated from c17.2 cells expressing cMyc-tagged Nestin constructs. Only NestinN1-640, NestinN1-1170, and NestinN1-1864 were found coimmunoprecipitated with c-Myc, indicating the Cdk5 binding area as NestinN1-640. Abbreviations: Cdk5, cyclin-dependent kinase 5; CP, cortical plate; DAPI, 4',6-diamidino-2-phenylindole; IP, immunoprecipitation; NSPC, neural stem/progenitor cell; NTC, non-targeting control; SVZ, subventricular zone.

confirm that Nestin expression is associated with Cdk5 shuttling between the cytosol and nucleus and, thus, alters cytosolic Cdk5 activity and Drp1 phosphorylation levels.

As a member of the IF protein family, we hypothesized that Nestin functions as a signaling scaffold in Cdk5 regulation through their direct interaction [12]. To define the potential Nestin-binding domain in Cdk5, we constructed different length plasmids of Cdk5 expressed as Flag fusion proteins and transiently transfected them into c17.2 cells (Fig. 5F). Cdk5N1–64 and Cdk5N1–147, as well as full-length Cdk5, co-immunoprecipitated with Nestin, which restricted Nestin binding to a specific region (1–64 amino acid) in Cdk5 (Fig. 5G). Furthermore, the immunofluorescence staining showed that Cdk5N1–64 localized almost exclusively in the nucleus, while Cdk5N1–147 and Cdk5 were distributed throughout the whole cell. Other constructs including Cdk5N65–147, Cdk5N148–292, and Cdk5N65–292 were expressed mostly in the cytosol after transfection into 293FT cells (Supporting Information Fig. S10). We conclude that Cdk5N1–64 is essential for the nuclear localization of Cdk5. Nestin binding with the Cdk5N1–64 residues may contribute to its role in the regulation of Cdk5 subcellular distribution. Additionally, we also synthesized different Nestin fragments and inserted them into a cMyc vector. Similarly, NestinN1–307, NestinN1–604, NestinN1–1170, NestinN641–1864, and NestinN1–1864 were overexpressed in c17.2 cells to find out the Cdk5 binding fragment on Nestin (Fig. 5H). Using the co-IP, NestinN1–640 was identified as the Cdk5-binding area (Fig. 5I). To validate the functional role of NestinN1–640, we transfected this plasmid into Nestin-knockdown cells and found that more Cdk5 was retained in the cytosol after the transfection (Fig. 6A). Correspondingly, NestinN1–640 overexpression also enhanced Drp1 phosphorylation and mitochondrial fragmentation with increased glycolysis production and diminished ATP generation (Fig. 6B–6E). Overexpression of NestinN1–640 also rescued the repressed self-renewal capacity of Nestin-knockdown cells (Fig. 6F). Together, these results provide a potential mechanism by which Nestin controls mitochondrial dynamics and the stemness of NSPCs.

DISCUSSION

Nestin is ubiquitously expressed in proliferating stem/progenitor cells and regenerating tissues, indicating that it may have some critical physiological functions. Here, we show that Nestin contributes to NSPC stemness by regulating the metabolism and dynamics of mitochondria via the mediation of Drp1 phosphorylation at serine 592. These results suggest that Nestin is not only a marker of multilineage stem/progenitor cells but also a potential key cellular signaling molecule in stemness maintenance.

Metabolic flexibility enables stem/progenitor cells to meet the divergent demands of their specialized functions, including self-renewal and differentiation. Stem/progenitor cells in the multipotent state preferentially rely on aerobic glycolysis to support anabolic metabolism, ensuring cell viability. In contrast, differentiated cells rely on OXPHOS [26]. However, the role of metabolic circuits in NSPCs maintenance is certainly a novel field. Zheng et al. demonstrated that key glycolytic enzymes including hexokinase and lactate dehydrogenase

were downregulated during the neuronal differentiation of NSPCs, marking the transition from glycolysis of NSPCs to neuronal OXPHOS [35], which was consistent with the mitochondrial metabolic transition pattern in our study. In addition, Stoll et al. found that mitochondrial quantity and OCRs were decreased accompanied with fewer new neurons produced by NSPCs with aging [36]. As a stem cell marker, knockdown of Nestin in NSPCs showed some characteristics of aging, such as declined proliferative capacity. The mitochondrial mass and metabolism transition were inconsistent with those of the aging NSPCs, suggesting Nestin-knockdown NSPCs were distinct from aging cells. Thus, the functions of Nestin in aging NSPCs need to be further explored.

On the other hand, many studies have demonstrated that lipid metabolism greases the stem cell engine, apart from glucose metabolism [37]. Knobloch et al. first figured out an essential role of fatty acid synthase-dependent lipogenesis in NSPCs proliferation [38]. Furthermore, Stoll et al. demonstrated that NSPCs within the adult subventricular zone expressed enzymes of fatty acid oxidation to sustain aerobic respiration. Inhibition of fatty acid oxidation decreased the proliferation of NSPCs [39]. Thus, concise metabolic regulation is important for controlling NSPC fate. Moreover, some researches focused the mitochondrial morphology and biogenesis on adult hippocampus neural stem cells (NSCs), making mitochondrial function a potential therapeutic target. Steib et al. found the development of new neurons from adult hippocampus NSCs under normal or exercise condition was paralleled with extensive mitochondrial biogenesis and increased dendritic mitochondrial contents, thus mitochondria are targets for enhancing neurogenesis-dependent hippocampal plasticity [40]. Besides that, Beckervordersandforth et al. found age-associated alterations in mitochondrial function and morphology in NSCs and suggested that pharmacological enhancement of mitochondrial function could ameliorate age-associated neurogenesis defects [41]. Here, we found that Nestin modulates NSPC stemness by affecting mitochondrial metabolism via a Nestin-Cdk5-Drp1 axis.

The impact of mitochondrial morphology on metabolism has been widely explored in several studies [42–44]. The mitochondrial network is generally mediated by mitochondrial dynamic proteins including Drp1, MFN1, MFN2, and OPA1. In this study, we reported that the expression of Drp1 (which mainly mediates the mitochondrial fission process) phosphorylation mutants could influence the fate of NSPCs, suggesting that Drp1 activation might play an important role in CNS development. Interestingly, Waterham et al. reported a newborn patient who harbored a dominant-negative mutation in Drp1 and exhibited microcephaly and abnormal brain development [45]. We previously observed that Nestin knockdown inhibited the mitochondrial recruitment of Drp1 and induced mitochondrial elongation in GIST cells [16], but we did not detect any obvious difference in the Drp1 distribution between Nestin-knockout and NTC cells (data not shown). In this study, we found that Nestin knockdown reduced Drp1 phosphorylation and, consequently its activity in NSPCs. Exogenous expression of the phospho-mimetic mutant, pDrp1^{S592E}, reversed the mitochondrial network and metabolic phenotype of Nestin-knockdown NSPCs and partially rescued their self-renewal capacity. We thus conclude that Drp1 activity plays a critical role in regulating NSPC stemness.

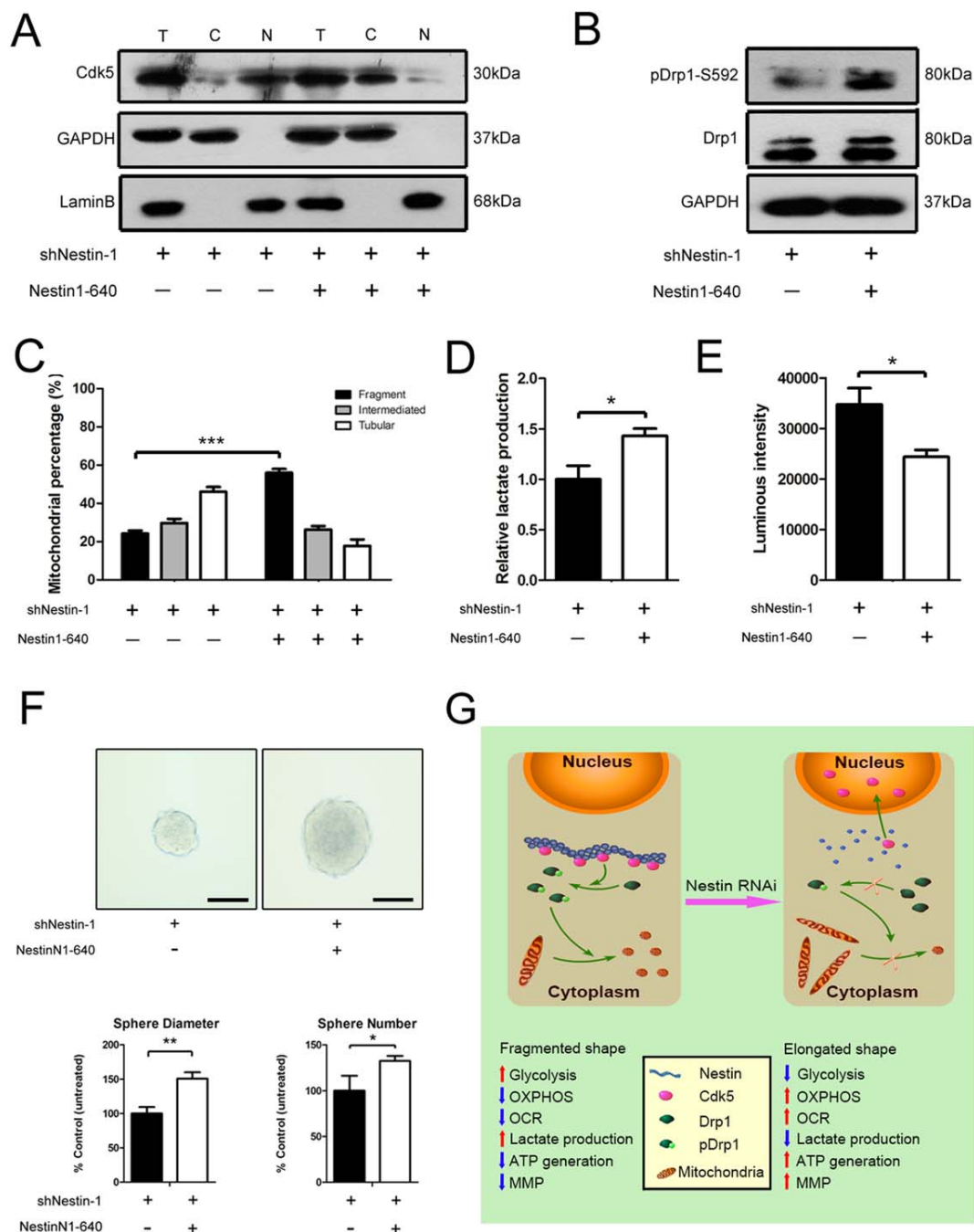


Figure 6. NestinN1-640 overexpression increases the sequestered Cdk5 in cytosol and promotes mitochondrial fragmentation and glycolysis. **(A, B)**: Transfection of NestinN1-640 induced Cdk5 redistribution (A) and subsequent Drp1 activation (B) in Nestin-knockdown cells. **(C)**: Statistical analysis of mitochondrial morphology alterations after NestinN1-640 overexpression. **(D, E)**: NestinN1-640 overexpression enhanced lactate production (D) and reduced ATP generation (E). **(F)**: Neurosphere-forming assays illustrated the effect of Nestin depletion on the self-renewal and proliferation of NSPCs. Scale bar = 100 μ m. Sphere diameters were calculated ($n = 5$) and the diameter was determined using the Image J software ($n = 5$). All data are expressed as means $\pm$ SD; *, $p < .05$; ***, $p < .001$, as assessed by t test. **(G)**: A model of Nestin as a metabolic mediator of Cdk5-Drp1 axis in NSPCs. Abbreviations: Cdk5, cyclin-dependent kinase 5; Drp1, dynamin-related protein 1; MMP, mitochondrial membrane potential; OCR, oxygen consumption rate; OXPHOS, oxidative phosphorylation; pDrp1, phosphorylated (activated) Drp1.

Drp1 activity is mediated by post-translational modifications, especially phosphorylation. To determine the molecular mechanism underlying Drp1 activation in NSPCs, we investigated the potential kinases implicated in Drp1 phosphorylation.

The Cdk family kinases, Cdk1 and Cdk5, have been reported to phosphorylate Serine 616 of human Drp1, which is analogous to Serine 592 in mouse Drp1. Cdk5 is a multi-faceted kinase that contributes to neuronal maturation and migration and thus,

participates in the development and pathological conditions of the CNS. Previous studies have shown that nuclear Cdk5 is an essential cell-cycle suppressor in the CNS. For example, Zhang et al. found that, following serum starvation, Cdk5 inhibited the reentry of neurons into the cell cycle by binding with nuclear-accumulated p27 [46]. Another anti-proliferative mechanism of nuclear Cdk5 is its interaction with the cell-cycle-associated transcription factor, E2F1, which disturbs the association of E2F1 with its cofactor, DP1, in post-mitotic neurons [47]. Additionally, Nestin has been reported to regulate the differentiation process through Cdk5 in myoblasts [48, 49]. However, the underlying mechanisms remain elusive. Here, we note that, in NSPCs, Nestin knockdown promotes the translocation of Cdk5 from the cytosol into the nucleus, reducing the level of cytosolic Cdk5 and the downstream activation of Drp1, thereby triggering mitochondrial elongation. Furthermore, the Co-IP studies provide convincing evidence of a direct interaction between Nestin and Cdk5. Based on these findings, we propose that cytosolic Cdk5 may regulate cellular metabolism and NSPC fate through its impact on mitochondrial dynamics. We further hypothesize that the respective decrease and increase in cytosolic and nuclear Cdk5 mutually give rise to the proliferation defects seen in Nestin-knockdown NSPCs. Additional research is warranted to investigate the physiological roles of nuclear Cdk5 in NSPCs.

Collectively, these findings offer fundamental insights into the relationship between the Drp1-mediated regulation of mitochondrial dynamics and the maintenance of NSPC stemness. Moreover, these data reveal that a Nestin-Cdk5-Drp1 axis is involved in modulating NSPC homeostasis, thereby uncovering a molecular function for Nestin.

CONCLUSION

Nestin sequesters the cytoplasmic localization of Cdk5, enhances Drp1 phosphorylation, mitochondrial fragmentation, and

glycolysis, which inhibits NSPC differentiation and maintains stemness.

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AUTHOR CONTRIBUTIONS

J.W., Y.H., and J.C.: conception and design, collection and/or assembly of data, manuscript writing; Q.K.: collection and/or assembly of data, data analysis and interpretation; J.X., W.H., H.W., Y.Z., X.S., and A.S.A.B.: data analysis and interpretation; H.L., Y.Q., Y.W., and B.Z.: collection and/or assembly of data; A.P.X.: conception and design, financial support, manuscript writing, final approval of manuscript.

DISCLOSURE OF POTENTIAL CONFLICTS OF INTEREST

The authors indicated no potential conflicts of interest.

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